HIDDEN COORDINATES AND SELECTION RULES IN THE FOURIER ENTROPY-INFLUENCE CONJECTURE

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Abstract. The erasure entropy $E_i(f)$ of a coordinate i of a Boolean function f is the entropy of the event $i \in U$ given $U \setminus \{i\}$, with U drawn from the spectral distribution of f. The Fourier entropy-influence conjecture is equivalent to every non-constant f having a coordinate whose erasure entropy is at most a constant times its influence. We give monotone functions with a hidden coordinate, which enters only through one point below a majority threshold, whose erasure entropy is unbounded relative to its influence, while the erasure entropies of all coordinates sum to at most twice the total influence. Hence no rule that chooses the coordinate by its influence, by its normalised level-two weight or by shortest minterms works, even for monotone functions. We also compute three extremal ratios over monotone functions of up to seven variables.

1. Introduction

For $f\colon \{0,1\}^n \to \{-1,1\}$ write $f = \sum_S \hat{f}(S)\chi_S$ with $\chi_S(x) = (-1)^{\sum_{j\in S} x_j}$, and let U be a random subset with $\Pr[U = S] = \mu(S) = \hat{f}(S)^2$. The Fourier entropy is $H(f) = H(U)$ and the total influence is $I(f) = \mathbb{E}|U| = \sum_i \mathrm{Inf}_i(f)$, where $\mathrm{Inf}_i(f) = \Pr[i \in U]$. The Fourier entropy-influence (FEI) conjecture of Friedgut and Kalai [4] asks for an absolute C with $H(f) \le C\, I(f)$ for every Boolean f. It is known for symmetric functions [17], for read-once formulas [16, 3] and for linear threshold functions [21]. It is also known for plateaued functions [18], for a class of cryptographic Boolean functions of logarithmic density one [6] and for read-k decision trees with k fixed [29]. Its min-entropy version is known for read-k DNFs (formulas in disjunctive normal form) with k fixed [19, 1], and Kelman, Kindler, Lifshitz, Minzer and Safra [9] prove that the min-entropy is $O(I(f)\log I(f))$ for functions of constant variance, with a matching concentration of the spectrum. Han [7] bounds $H(f)$ by a multiple of $I(f)$ plus $\sum_i \mathrm{Inf}_i(f)\log_2(1/\mathrm{Inf}_i(f))$, and Li and Han [15] lower the multiple to the sharp value $1/\ln 2$. Hod [8] shows, with monotone functions, that the best constant exceeds 6.45, and explicit balanced monotone functions on 14 to 18 variables raise this to 6.5218 [11, 12, 13].

The erasure entropy of f at i is

$$E_i(f) = H\big(\mathbf{1}[i \in U] \,\big|\, U \setminus \{i\}\big),$$

the conditional entropy of one coordinate given the others in the sense of Verdú and Weissman [28]. It is the part of the entropy drop from f to its two restrictions at x_i that is not controlled by a constant times Inf_i [7, Remarks 3.1–3.2], [5, (32)], and the FEI conjecture holds if and only if

$$G = \sup_f \min_{i\colon \mathrm{Inf}_i(f)>0} \frac{E_i(f)}{\mathrm{Inf}_i(f)}$$

is finite [20]. One direction is immediate from $\sum_i E_i \le H$. The other restricts f one coordinate at a time, each time at a coordinate with small ratio, and does not say which coordinate that is.

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This paper is about that choice. For monotone linear DNFs of large width in which two terms share at most one variable, every relevant coordinate has $E_i/\mathrm{Inf}_i = 2 + o(1)$ [24], so there the choice does not matter; an upper bound on each E_i obtained by revealing the other coordinates one at a time is given in [23].

Section 3 studies the functions

$$f_k(y, x) = \mathrm{MAJ}_k(y) \vee \big(x \wedge [y \geq y^0]\big), \qquad k \text{ odd},$$

where $y^0 \in \{0,1\}^k$ has $(k-1)/2$ ones and $[y \geq y^0]$ is the AND (conjunction) of the coordinates of y at which y^0 is 1. The variable x changes the value only at the single point $y = y^0$, so $\mathrm{Inf}_x(f_k) = 2^{-k}$, and the ratio E_x/Inf_x tends to infinity (Proposition 5). Nevertheless the erasure entropies of all coordinates together are small:

Theorem 1. *For every odd k, $\sum_i E_i(f_k) \leq 2\, I(f_k)$. If every y_j is replaced by a non-constant self-dual monotone function φ_j of its own block of fresh variables, the resulting monotone function F satisfies $E_x(F) = E_x(f_k)$, $E_\ell(F) < 4\,\mathrm{Inf}_\ell(F)$ at every relevant variable $\ell \neq x$, and $\sum_i E_i(F) \leq 5\, I(F)$.*

The large ratio at x is paid for by the total influence: $E_x \leq h_2(2^{-k})$, which is far below $I(f_k) \geq 1$. Two facts drive the proof. The parity of $|U|$ determines whether $i \in U$ once $U \setminus \{i\}$ is known, so every E_i is at most the binary entropy of the Fourier weight on even levels, and for f_k that weight is exactly 2^{-k} (Lemma 3). After the substitution, E_ℓ has an exact expression in terms of the spectrum of f_k (Proposition 6). Theorem 1 does not by itself give $H \leq C\, I$ for these functions, since the passage from erasure entropies to H uses a good coordinate in every restriction and the family is not closed under restrictions; its point is that an unbounded ratio at one coordinate is compatible with a bounded sum.

Section 5 uses these functions against simple selection rules.

Theorem 2. *For each of the following rules there are monotone functions on which the coordinate it selects has E_i/Inf_i as large as we like, with ties broken in any way: the coordinate of smallest influence; of largest influence; of largest normalised level-two weight $W_i^{(2)}/\mathrm{Inf}_i$; of smallest normalised level-two weight; the variable lying in the most minterms of smallest size; the variable of smallest E_i/Inf_i among those lying both in a minterm of smallest size and in a maxterm of smallest size.*

Here $W_i^{(2)}(f) = \sum_{j \neq i} \hat{f}(\{i,j\})^2$. That the largest influence fails was shown in [22] by a different construction. For the rule of smallest normalised level-two weight, f_k itself is not a counterexample; there the hidden coordinate is made pivotal at two complementary points of the middle layer, which removes its level-two weight altogether (Proposition 9).

Section 6 shows that the identities $\hat{f}(\{i\})^2 = \mathrm{Inf}_i(f)^2$ and $\sum_i \mathrm{Inf}_i(f)^2 \leq 1$, which hold for monotone f, are not enough to bound the erasure entropy: an exchangeable probability law on subsets of $[n]$ satisfies both and has $E_i/\Pr[i \in U] \geq \log_2(n)/288$. Section 7 gives the exact maxima of the three ratios $\min_i E_i/\mathrm{Inf}_i$, H/I and $\sum_i E_i/I$ over all monotone functions on at most seven variables, and the worst cases of the simple rules on at most six.

The last rule of Theorem 2 needs more than the hidden coordinate of f_k, which lies in no shortest maxterm. The rule is bounded by 2.6974 when a shortest minterm or maxterm is a single variable (Proposition 10), and on at most six variables its worst case equals $G_{\mathrm{mon}}(n)$. It fails because a variant of f_k has its hidden coordinate in both a shortest minterm and a shortest maxterm (Proposition 11), and substituting threshold functions of slightly different minterm and maxterm sizes leaves that coordinate the only variable in both, with almost the same ratio (Proposition 12).

2. Preliminaries

All logarithms are to base 2, $\eta = \log_2 e$, and h_2 is the binary entropy. For $p, q \ge 0$ put

$$\Phi(p,q) = (p+q)\, h_2\Big(\frac{q}{p+q}\Big) = p\log\frac{p+q}{p} + q\log\frac{p+q}{q}, \qquad \Phi(p,0) = \Phi(0,q) = 0.$$

Then

$$E_i(f) = \sum_{T \not\ni i} \Phi\big(\mu(T), \mu(T\cup\{i\})\big), \qquad \mathrm{Inf}_i(f) = \sum_{T\not\ni i} \mu(T\cup\{i\}). \tag{1}$$

Φ is concave and positively homogeneous of degree one, hence superadditive: $\sum_t \Phi(p_t,q_t) \le \Phi(\sum_t p_t, \sum_t q_t)$. It is nondecreasing in each argument, and $\Phi(p,q) \ge q\log(p/q)$ when $p \ge q$, since both of its terms are nonnegative and $(p+q)/q \ge p/q$. We use $h_2(t) \le t\log(e/t)$, which follows from $-(1-t)\ln(1-t) \le t$.

For monotone f (true may be written -1 or $+1$; squared coefficients do not depend on the choice), $|\hat f(\{i\})| = \mathrm{Inf}_i(f)$. A Boolean function is self-dual if $f(\bar x) = -f(x)$, where $\bar x$ is the complement of x; then $\hat f(S) = 0$ for every S of even size, and in particular $\hat f(\emptyset) = 0$.

Lemma 3. *Let $\varepsilon = \Pr[\,|U| \text{ even}]$. Then $E_i(f) \le h_2(\varepsilon)$ for every i, and $E_i(f) \le h_2(\mathrm{Inf}_i(f))$.*

Proof. Given $U\setminus\{i\}$, the parity of $|U|$ and the indicator of $i \in U$ determine each other, so $E_i = H(|U| \bmod 2 \mid U\setminus\{i\}) \le H(|U| \bmod 2) = h_2(\varepsilon)$. The second bound drops the conditioning. □

3. The hidden coordinate

Fix an odd k, let $y^0 \in \{0,1\}^k$ have $(k-1)/2$ ones, and let f_k be as in the introduction, written in ±1 values with $+1$ for true. Put

$$\delta = 2^{-k}, \qquad a = \binom{k-1}{(k-1)/2} 2^{1-k},$$

so that a is the influence of every variable of MAJ_k. For $k = 1$ the vector y^0 is empty and $f_1 = y \vee x$.

Lemma 4. (1) *$f_k = \mathrm{MAJ}_k + (1+x)\mathbf{1}[y = y^0]$, where x is the ±1 value of the hidden variable.*
(2) *$\mathrm{Inf}_x(f_k) = \delta$; $\mathrm{Inf}_{y_j}(f_k) = a+\delta$ if $y^0_j = 1$ and $a-\delta$ if $y^0_j = 0$; $I(f_k) = ka \ge 1$.*
(3) *The Fourier weight of f_k on sets of even size is exactly δ.*

Proof. If $y \ge y^0$ and $y \ne y^0$ then $|y| \ge (k+1)/2$, so the clause $x \wedge [y \ge y^0]$ changes the value only at $y = y^0$, where $\mathrm{MAJ}_k = -1$ and $f_k = x$; this is (1). The derivative in x is $\mathbf{1}[y = y^0]$, so $\mathrm{Inf}_x = \Pr[y = y^0] = \delta$. Setting $f_k(1, y^0) = 1$ creates a pivotal edge in direction j when $y^0_j = 1$, since $\mathrm{MAJ}_k = -1$ at the neighbour of y^0 in direction j, and removes one when $y^0_j = 0$, since $\mathrm{MAJ}_k = 1$ at that neighbour; each edge carries influence δ. There are $(k-1)/2$ coordinates of the first kind and $(k+1)/2$ of the second, so $I(f_k) = \delta + ka - \delta = ka$. As a is the largest atom of the binomial law on $\{0,\dots,k-1\}$, $a \ge 1/k$. For (3): MAJ_k is self-dual, so its spectrum lies on odd sets not containing x. The coefficients of $(1+x)\mathbf{1}[y = y^0]$ all have modulus δ, one at each T and one at each $T\cup\{x\}$, $T \subseteq [k]$; the 2^k sets of even size among them carry no weight of MAJ_k and have total weight $2^k\delta^2 = \delta$. □

Proof of Theorem 1 for f_k. By Lemmas 3 and 4, $\sum_i E_i(f_k) \le (k+1)h_2(\delta)$, and $I(f_k) \ge 1$. For $k = 1$, $f_1 = y \vee x$ has μ uniform on four sets, so $E_x = E_y = 1$ and $\sum_i E_i = 2 = 2I$. For $k = 3$, $I = 3/2$ and $4h_2(1/8) < 2.18 < 3$. For odd $k \ge 5$, $(k+1)h_2(\delta) \le (k+1)(k+\eta)2^{-k}$, which is < 1.21 at $k = 5$ and decreases, since its ratio from k to $k+2$ is $\frac14 \frac{k+3}{k+1}\frac{k+2+\eta}{k+\eta} < 1$. □

Proposition 5. *For odd k large enough,*

$$\frac{E_x(f_k)}{\mathrm{Inf}_x(f_k)} \;\geq\; \Big(\frac{1}{2} - \frac{4k}{(k-3)^2}\Big)\Big(\frac{k-1}{4} - 2\log k\Big).$$

Proof. Write $g = \mathrm{MAJ}_k + \mathbf{1}[y = y^0]$ for the average of f_k over x. By (1), $E_x = \sum_T \Phi(\hat{g}(T)^2, \delta^2)$ over $T \subseteq [k]$, and $\mathrm{Inf}_x = 2^k\delta^2$, so E_x/Inf_x is the average of $\delta^{-2}\Phi(\hat{g}(T)^2, \delta^2)$ over uniform T. For $|T| = r$ odd, $\widehat{\mathrm{MAJ}_k}(T)$ has modulus

$$b_r = a\binom{(k-1)/2}{(r-1)/2}\Big/\binom{k-1}{r-1}.$$

Indeed, for $j \in T$ it equals the coefficient at $T \setminus \{j\}$ of the derivative in y_j, the indicator of the middle layer of the other $2m = k-1$ coordinates, evaluated at a uniform point v of $\{0,1\}^{2m}$, and $\sum_S \mathbb{E}[\chi_S \mid |v| = m]\,\xi^{|S|} = (1-\xi)^m(1+\xi)^m = (1-\xi^2)^m$ gives $\mathbb{E}[\chi_S \mid |v| = m] = (-1)^s\binom{m}{s}/\binom{2m}{2s}$ for $|S| = 2s$. If $(k-1)/4 \leq r-1 \leq 3(k-1)/4$ then $\binom{m}{(r-1)/2} \geq 2^{(k-1)/8}$, and with $a \geq 1/k$ and $\binom{k-1}{r-1} \leq 2^{k-1}$ we get $b_r/\delta \geq 2^{(k-1)/8+1}/k \geq 2$ for large k. Then $|\hat{g}(T)| \geq b_r - \delta \geq b_r/2$ and $\delta^{-2}\Phi(\hat{g}(T)^2, \delta^2) \geq \log(\hat{g}(T)^2/\delta^2) \geq (k-1)/4 - 2\log k$. The set of such T has probability at least $\frac{1}{2} - \Pr[\,|\,|T| - k/2| > (k-3)/4\,] \geq \frac{1}{2} - \frac{4k}{(k-3)^2}$ by Chebyshev's inequality. □

The ratio is 3.52, 5.78 and 9.27 at $k = 3, 9, 17$.

4. Substitution

Let $\varphi_1, \ldots, \varphi_k$ be non-constant self-dual monotone functions on disjoint blocks of fresh variables, and $F(z, x) = f_k(\varphi_1(z^1), \ldots, \varphi_k(z^k), x)$. Write μ for the spectral distribution of f_k and ι_j for $\mathrm{Inf}_{y_j}(f_k)$. For a variable ℓ of block j let $\tau = \mathrm{Inf}_\ell(\varphi_j) = |\hat{\varphi}_j(\{\ell\})|$.

Proposition 6. *With $p_T = \mu(T)$ and $q_T = \mu(T \cup \{y_j\})$ for T a subset of the coordinates of f_k other than y_j,*

$$\mathrm{Inf}_\ell(F) = \tau\iota_j, \qquad E_\ell(F) = \sum_T \Phi(p_T, \tau^2 q_T).$$

Moreover $E_x(F) = E_x(f_k)$, $\mathrm{Inf}_x(F) = \delta$, and the weight of F on sets of even size is δ.

Proof. Since $\hat{\varphi}_j(\emptyset) = 0$, expanding $f_k = \sum_T \hat{f}_k(T)\prod_{j\in T} y_j$ and substituting gives $\hat{F}(S) = \hat{f}_k(T)\prod_{j\in T}\hat{\varphi}_j(S_j)$, where $S_j = S \cap \mathrm{block}_j$ and T consists of x if $x \in S$ and the j with $S_j \neq \emptyset$. Fix the parts of a set $V \not\ni \ell$ outside block j. If $V_j \neq \emptyset$, the sets V and $V \cup \{\ell\}$ have the same T, and their block-j parts differ in size by one; as φ_j is self-dual, one of the two coefficients vanishes and the pair contributes $\Phi(\cdot, 0) = 0$. If $V_j = \emptyset$, the pair is $(T, T \cup \{y_j\})$ with factors 1 and τ^2, times a common factor $\prod_{i\in T}\hat{\varphi}_i(S_i)^2$. By homogeneity of Φ and $\sum_{S_i\neq\emptyset}\hat{\varphi}_i(S_i)^2 = 1$ (Parseval and $\hat{\varphi}_i(\emptyset) = 0$), summing over the other blocks leaves $\Phi(p_T, \tau^2 q_T)$. The influence follows in the same way from $\sum_{S\ni\ell}\hat{\varphi}_j(S)^2 = \tau$. For x, the pairs $(V, V \cup \{x\})$ correspond to $(T, T \cup \{x\})$ with a common factor, so E_x and Inf_x are unchanged. Each nonempty S_i has odd size, so $|S| \equiv |T| \pmod 2$ and the even weight is that of f_k. □

Corollary 7. $E_\ell(F) \leq \min\big(h_2(\delta), h_2(\tau^2\iota_j)\big)$.

Proof. The first bound is Lemma 3 with Proposition 6. For the second, superadditivity gives $E_\ell(F) \leq \Phi(\sum_T p_T, \tau^2\sum_T q_T) = \Phi(1-\iota_j, \tau^2\iota_j) \leq \Phi(1-\tau^2\iota_j, \tau^2\iota_j)$, using that Φ is nondecreasing in its first argument. □

Proof of Theorem 1 for F. Let $k \geq 3$. Then $\delta \leq 1/(2k)$, so $\iota_j \geq a - \delta \geq 1/(2k)$. Put $\tau_0 = \sqrt{\delta/\iota_j}$. If $\tau \geq \tau_0$, Corollary 7 and $h_2(t) \leq t \log(e/t)$ give

$$\frac{E_\ell}{\mathrm{Inf}_\ell} \leq \frac{h_2(\delta)}{\tau_0 \iota_j} \leq \tau_0 \log \frac{e}{\delta}.$$

If $\tau \leq \tau_0$, they give $E_\ell/\mathrm{Inf}_\ell \leq \tau \log(e/(\tau^2 \iota_j))$, whose derivative in τ is $\log(1/(\tau^2\iota_j)) - \eta > 0$, because $\tau^2 \iota_j \leq \delta \leq 1/8 < 1/e$; so it is at most its value $\tau_0 \log(e/\delta)$ at $\tau = \tau_0$. Hence

$$\frac{E_\ell(F)}{\mathrm{Inf}_\ell(F)} \leq \sqrt{\delta/\iota_j}\, \log(e/\delta) \leq \gamma_k := \sqrt{2k2^{-k}}\,(k+\eta).$$

Here $\gamma_3 < \frac{9}{4}\sqrt{3} < 4$, and $\gamma_{k+2}/\gamma_k = \frac{1}{2}\sqrt{1+2/k}\,(1+2/(k+\eta)) \leq \frac{1}{2}\sqrt{5/3} \cdot \frac{3}{2} < 1$. For $k = 1$, $\delta = \iota_1 = 1/2$ and $E_\ell/\mathrm{Inf}_\ell \leq h_2(\tau^2/2)/(\tau/2) \leq \tau \log(2e/\tau^2) \leq 2\sqrt{2/e}\,\eta < 3$.

For the sum, $I(F) = \delta + \sum_j \iota_j I(\varphi_j)$, and $I(\varphi_j) \geq \mathrm{Var}(\varphi_j) = 1$, so $I(F) \geq \delta + \sum_j \iota_j = I(f_k) \geq 1$. Therefore

$$\sum_i E_i(F) = E_x(F) + \sum_\ell E_\ell(F) \leq h_2(\delta) + 4\sum_j \iota_j I(\varphi_j) \leq I(F) + 4I(F). \qquad \square$$

Remark 8. The bound $E_\ell < 4\mathrm{Inf}_\ell$ holds at every variable of every block, whatever the sizes of the blocks; in particular for majority blocks of any size and for iterated majorities. Since x keeps the same E_x and Inf_x, the ratio at x still tends to infinity with k.

5. Selection rules

Proof of Theorem 2. Smallest influence. In f_k, $\mathrm{Inf}_x = \delta < a - \delta \leq \mathrm{Inf}_{y_j}$ for $k \geq 3$, so the rule picks x, and Proposition 5 applies.

Largest normalised level-two weight. MAJ_k has no weight on sets of size two, so every coefficient of f_k at a set of size two is a coefficient of $(1+x)\mathbf{1}[y = y^0]$, of modulus δ. Thus $W_i^{(2)} = k\delta^2$ for every coordinate, and maximising $W_i^{(2)}/\mathrm{Inf}_i$ is minimising Inf_i.

Largest influence. Substitute $\varphi_j = \mathrm{MAJ}_{2w+1}$ for every j. Then $\tau = \binom{2w}{w}4^{-w} \leq (w+1)^{-1/2}$ (the ratio from w to $w+1$ is $\frac{2w+1}{2w+2}$, and $(\frac{2w+1}{2w+2})^2 \leq \frac{w+1}{w+2}$). For $w+1 > 4^k$ every variable of a block has influence $\tau\iota_j < \delta$, so x is the unique coordinate of largest influence, and by Proposition 6 its ratio is that of f_k.

Most shortest minterms. Substitute $\varphi_j = \mathrm{MAJ}_3$. The minterms of F are x together with a minterm of MAJ_3 in each block j with $y_j^0 = 1$, of size $1 + 2 \cdot \frac{k-1}{2} = k$, and minterms of MAJ_3 in $(k+1)/2$ of the blocks, of size $k+1$. So every shortest minterm contains x, while a variable of a block lies in two of the three minterms of MAJ_3 and hence in fewer of them. The rule picks x.

Smallest normalised level-two weight. This is Proposition 9 below.

Smallest ratio among the variables in a shortest minterm and a shortest maxterm. This is Proposition 12 below. $\square$

For f_k the hidden coordinate has the same level-two weight as every other coordinate, which is not the smallest normalised value. The next construction removes it.

Let $k \geq 6$ be even, let $y^0 \in \{0,1\}^k$ have $k/2$ ones. Let $N(y) = 1$ if $|y| \geq k/2$ and 0 otherwise, and define $f'_k(y,x) = N(y)$ except that $f'_k(y,x) = x$ when $y \in \{y^0, \bar{y}^0\}$. Lowering a point of the middle layer from 1 to 0 preserves monotonicity, so f'_k is monotone.

Proposition 9. *$W_x^{(2)}(f'_k) = 0$ and $W_{y_j}^{(2)}(f'_k) > 0$ for every j, so x is the unique coordinate of smallest normalised level-two weight;* $\mathrm{Inf}_x = 2^{1-k}$, $\mathrm{Inf}_{y_j} = \binom{k}{k/2}2^{-k}$ *for every j; and for even k*

large enough

$$\frac{E_x(f'_k)}{\mathrm{Inf}_x(f'_k)} \;\ge\; \Big(1-\frac{8}{k}\Big)\Big(\frac{k}{4}-4-2\log(k+1)\Big).$$

Proof. With $D=\mathbf{1}[y\in\{y^0,\bar y^0\}]$ we have $f'_k=N-(1-x)D$ in ±1 values, the derivative in x is D, and the average over x is $g=N-D$. Put $\delta'=2^{1-k}$. Since $\chi_T(\bar y^0)=(-1)^{|T|}\chi_T(y^0)$, $\hat D(T)$ vanishes for odd $|T|$ and has modulus δ' for even $|T|$. So $\mathrm{Inf}_x=\Pr[y\in\{y^0,\bar y^0\}]=\delta'$ and the coefficient of f'_k at $\{x,y_j\}$, which is $\hat D(\{y_j\})$, is zero: $W^{(2)}_x=0$. At y^0 and at $\bar y^0$ the change creates or removes a pivotal edge in direction j according to the jth bit, and the two bits are opposite, so $\mathrm{Inf}_{y_j}(f'_k)=\mathrm{Inf}_j(N)=\binom{k}{k/2}2^{-k}=:a_0$.

In ±1 values, N is the sum of an odd function, the sign of $|y|-k/2$ with 0 on the middle layer, and the indicator of the middle layer. Only the latter has even coefficients, and by the generating function of Proposition 5, for even $r=|T|$,

$$|\hat N(T)| = a_0\binom{k/2}{r/2}\Big/\binom{k}{r}. \tag{2}$$

At $r=2$ this is $a_0/(k-1)$, and $a_0/((k-1)\delta')=\binom{k}{k/2}/(2(k-1))$ equals 2 at $k=6$ and increases. So $|\hat f'_k(\{y_j,y_{j'}\})|\ge|\hat N(\{y_j,y_{j'}\})|-\delta'>0$ and $W^{(2)}_{y_j}>0$.

The derivative spectrum is uniform over the even subsets T of $[k]$, and E_x/Inf_x is the average of $\delta'^{-2}\Phi(\hat g(T)^2,\delta'^2)$. For even $r\in[k/4,3k/4]$, (2) with $\binom{k/2}{r/2}\ge 2^{k/8}$, $a_0\ge 1/(k+1)$ and $\binom{k}{r}\le 2^k$ gives $|\hat N(T)|\ge 2^{-7k/8}/(k+1)\ge 2\delta'$ for large k, hence $|\hat g(T)|\ge|\hat N(T)|/2$ and $\delta'^{-2}\Phi(\hat g(T)^2,\delta'^2)\ge k/4-4-2\log(k+1)$. By Chebyshev's inequality, conditioned on $|T|$ even, $|T|\in[k/4,3k/4]$ with probability at least $1-8/k$. □

At $k=18$ the ratio at x is 14.28, while the smallest ratio over all coordinates is 1.60.

Shortest minterms and shortest maxterms. In each of these functions the hidden coordinate does not lie both in a shortest minterm and in a shortest maxterm: in f_k and its substitutions it lies in no shortest maxterm, and in f'_k in no shortest minterm. For monotone f let $A(f)$ be the set of variables lying in some shortest minterm and $B(f)$ the set lying in some shortest maxterm. Every minterm meets every maxterm, so $A(f)\cap B(f)$ is not empty. Consider the rule that chooses a variable of $A(f)\cap B(f)$ minimising E_i/Inf_i. It is bounded when a shortest minterm or a shortest maxterm is a single variable, but not in general.

Proposition 10. *Let $f=x\wedge h$, where h is a monotone function of the other variables that is not identically false, and let $p=\Pr[h\ true]$. Then $\mathrm{Inf}_x(f)=p$ and*

$$\frac{E_x(f)}{\mathrm{Inf}_x(f)} = 2(1-p)+\frac{\Phi\big((1-p)^2,p^2\big)}{p} \;\le\; 4(1-p).$$

The middle expression tends to 2 as $p\to0$, equals 2 at $p=1/2$, and its maximum over $p\in(0,1]$ is 2.697377, attained near $p=0.1994$.

Proof. Write x and h in ±1 values with -1 for true. Then $f=\frac{1+h}{2}+x\frac{1-h}{2}$, so for T not containing x, $\hat f(T)=\frac12(\mathbf{1}[T=\emptyset]+\hat h(T))$ and $\hat f(T\cup\{x\})=\frac12(\mathbf{1}[T=\emptyset]-\hat h(T))$. As $\hat h(\emptyset)=1-2p$, we get $\mu(\emptyset)=(1-p)^2$, $\mu(\{x\})=p^2$, and $\mu(T)=\mu(T\cup\{x\})=\hat h(T)^2/4$ for $T\ne\emptyset$. Since $\Phi(t,t)=2t$, (1) gives

$$E_x=\Phi\big((1-p)^2,p^2\big)+\tfrac12\big(1-(1-2p)^2\big),\qquad \mathrm{Inf}_x=p^2+\tfrac14\big(1-(1-2p)^2\big)=p,$$

and the second term of E_x is $2p(1-p)$. For the bound, $\Phi(a,b)=(a+b)h_2(b/(a+b))\le 2\sqrt{ab}$ by the inequality $h_2(t)\le 2\sqrt{t(1-t)}$, which follows from $h_2(t)\le(4t(1-t))^{1/\ln 4}$ [27]; here $2\sqrt{ab}=2p(1-p)$. □

If a shortest maxterm of a monotone f is a single variable x, then $f = x \wedge h$ with h the restriction $x = 1$, and every shortest maxterm is a single variable of this kind. So every variable of $B(f)$, and hence of $A(f) \cap B(f)$, has $E_i/\mathrm{Inf}_i < 2.6974$. The dual $f^\dagger$, the negation of f at the complement of its argument, exchanges minterms and maxterms and has the same squared Fourier coefficients, so the same holds when a shortest minterm is a single variable. In these cases any variable of $A \cap B$ will do.

In general a variable of $A \cap B$ can be bad. For odd $k \geq 3$ let $m = (k-1)/2$, let y^0 have ones exactly at $y_1, \ldots, y_m$, let y^1 differ from y^0 only in $y^1_{m+1} = 1$, let M be MAJ_k with its value at y^1 changed to false, and

$$\tilde{f}_k(y, x) = M(y) \vee \big(x \wedge [y \geq y^0]\big).$$

The point y^1 lies in the lowest true layer of MAJ_k, so M and $\tilde{f}_k$ are monotone.

Proposition 11. *The variable x lies in a shortest minterm and in a shortest maxterm of $\tilde{f}_k$, $\mathrm{Inf}_x(\tilde{f}_k) = 2^{1-k}$, and for odd k large enough*

$$\frac{E_x(\tilde{f}_k)}{\mathrm{Inf}_x(\tilde{f}_k)} \;\geq\; \Big(\frac{1}{2} - \frac{4(k-1)}{(k-5)^2}\Big)\Big(\frac{k-1}{4} - 2\log k\Big).$$

Proof. A true point has weight at least $m+1$: with x false it needs $M(y)$, so $|y| \geq m+1$, and with x true it needs $y \geq y^0$ or $M(y)$. Hence $\{x, y_1, \ldots, y_m\}$ is a shortest minterm. Every point of weight $m+2$ is true: with x false, $|y| = m+2$ and $M(y)$ holds; with x true, $|y| = m+1$ and either $y = y^1 \geq y^0$ or $M(y)$ holds. The point with x false and $y = y^1$ is false and has weight $m+1$, so the variables at which it is 0, among them x, form a shortest maxterm.

In ± 1 values with $+1$ for true, $\tilde{f}_k = \mathrm{MAJ}_k + (1+x)\mathbf{1}[y = y^0] - (1-x)\mathbf{1}[y = y^1]$. The derivative in x is $D = \mathbf{1}[y = y^0] + \mathbf{1}[y = y^1]$ and the average over x is $g = \mathrm{MAJ}_k + \mathbf{1}[y = y^0] - \mathbf{1}[y = y^1]$. With $\delta = 2^{-k}$, $\hat{D}(T) = \delta(\chi_T(y^0) + \chi_T(y^1))$ vanishes if $y_{m+1} \in T$ and has modulus 2δ otherwise. So $\mathrm{Inf}_x = 2^{k-1}(2\delta)^2 = 2^{1-k}$, and E_x/Inf_x is the average of $(2\delta)^{-2}\Phi(\hat{g}(T)^2, 4\delta^2)$ over the 2^{k-1} sets $T \subseteq [k] \setminus \{m+1\}$. For these T, $\chi_T(y^0) = \chi_T(y^1)$, the two indicators cancel, and $|\hat{g}(T)| = b_r$ for $|T| = r$ odd, with b_r as in the proof of Proposition 5. If $(k-1)/4 \leq r-1 \leq 3(k-1)/4$, that proof gives $b_r/(2\delta) \geq 2^{(k-1)/8}/k \geq 1$ for large k, hence $(2\delta)^{-2}\Phi(b_r^2, 4\delta^2) \geq \log(b_r^2/(4\delta^2)) \geq (k-1)/4 - 2\log k$. Now $|T|$ is binomial with parameters $k-1$ and $\frac{1}{2}$; it is odd with probability $\frac{1}{2}$, and by Chebyshev's inequality $\big||T| - \frac{k-1}{2}\big| > \frac{k-5}{4}$ with probability at most $4(k-1)/(k-5)^2$. Otherwise $(k+3)/4 \leq |T| \leq (3k-7)/4$, which is the required range. □

Both factors of the bound in Proposition 11, and of those in Propositions 5 and 9, are positive only from $k = 45$, $k = 45$ and $k = 66$ respectively. For smaller k the displayed product says nothing: at $k = 5$ the right side of Proposition 5 is 16.4, a product of two negative factors.

At $k = 3, 5, \ldots, 13$ the ratio at x is 1.80, 2.56, 3.17, 3.90, 4.70 and 5.55, while the smallest ratio over $A(\tilde{f}_k) \cap B(\tilde{f}_k)$ is the smallest over all coordinates, 1.00 at $k = 3$ and 0.0060 at $k = 13$. So $\tilde{f}_k$ does not defeat the rule. A substitution removes the other variables from $A \cap B$.

For $t \geq 1$ let Th_t be the function of $2t$ variables that is true when at least t of them are true, and Th'_t the one that is true when at least $t+1$ are. Their shortest minterms have sizes t and $t+1$, and their shortest maxterms sizes $t+1$ and t. Let $F_{k,t}$ be obtained from $\tilde{f}_k$ by replacing y_j with Th_t of a block of fresh variables for $j \leq m$ and with Th'_t of a block of fresh variables for $j > m$.

Proposition 12. *$A(F_{k,t}) \cap B(F_{k,t}) = \{x\}$, and for each k,*

$$\frac{E_x(F_{k,t})}{\mathrm{Inf}_x(F_{k,t})} \to \frac{E_x(\tilde{f}_k)}{\mathrm{Inf}_x(\tilde{f}_k)} \qquad (t \to \infty).$$

Hence the rule that chooses the variable of smallest E_i/Inf_i in $A \cap B$ has unbounded ratio on monotone functions.

Proof. If each y_j of $\tilde{f}_k$ is replaced by a non-constant monotone φ_j of its own block, the minterms of the result are the sets $\bigcup_{j\in S} Q_j$, together with x when $x \in S$, where S is a minterm of $\tilde{f}_k$ and each Q_j is a minterm of φ_j; the same holds for maxterms. So a shortest minterm of the result comes from a minterm S of $\tilde{f}_k$ minimising $\sum_{j\in S} \alpha_j$, where $\alpha_x = 1$ and α_j is the size of a shortest minterm of φ_j, and likewise for maxterms, with the sizes $\beta_x = 1$ and β_j of shortest maxterms.

Write $Y_0 = \{y_1, \dots, y_m\}$ and $Y_1 = \{y_{m+2}, \dots, y_k\}$. The minterms of $\tilde{f}_k$ are $\{x\} \cup Y_0$ and the sets of $m+1$ variables y_j other than $Y_0 \cup \{y_{m+1}\}$: a true point either has $y \geq y^0$ and x true or satisfies M, and M is true exactly at the points of weight at least $m+1$ other than y^1. The first has $\sum \alpha_j = 1 + mt$. Any other contains $i \leq m$ variables of Y_0 and $m+1-i \geq 1$ others, and has $\sum \alpha_j = (m+1)t + (m+1-i) > 1 + mt$. The maximal false points of $\tilde{f}_k$ are the point with x false and $y = y^1$, and the points with x true and $|y| = m$, $y \neq y^0$; the maxterms are therefore $\{x\} \cup Y_1$ and the sets of $m+1$ variables y_j other than $Y_1 \cup \{y_{m+1}\}$. The first has $\sum \beta_j = 1 + mt$, and every other contains a variable of Y_0 and has $\sum \beta_j \geq (m+1)t + 1$. Hence every shortest minterm of $F_{k,t}$ consists of x and variables of the blocks of Y_0, every shortest maxterm consists of x and variables of the blocks of Y_1, and $A \cap B = \{x\}$.

For the ratio let π_j be the probability that φ_j is true, $\sigma_j^2 = \pi_j(1-\pi_j)$, and expand $\tilde{f}_k$, in ± 1 values, in the orthonormal basis $\prod_{j\in S}(y_j - \pi_j)/\sigma_j$ (times x when $x \in S$) of the product measure with $\Pr[y_j = 1] = \pi_j$ and x uniform (the biased setting of [10]); call the coefficients $c_\pi(S)$. Each $\psi_j = (\varphi_j - \pi_j)/\sigma_j$ has mean 0 and variance 1 under the uniform measure on its block, so $\sum_{R_j \neq \emptyset} \hat{\psi}_j(R_j)^2 = 1$, and the coefficient of $F_{k,t}$ at the union of $S \cap \{x\}$ and sets $R_j \neq \emptyset$, $j \in S \setminus \{x\}$, is $c_\pi(S) \prod_j \hat{\psi}_j(R_j)$. So the spectral sample of $F_{k,t}$ is obtained by drawing S with probability $c_\pi(S)^2$ and then each R_j independently. It contains x exactly when S does, it determines $S \setminus \{x\}$, and given $S \setminus \{x\}$ the R_j do not depend on whether $x \in S$. Hence $E_x(F_{k,t})$ and $\mathrm{Inf}_x(F_{k,t})$ are the erasure entropy and the influence of x for the law $c_\pi(S)^2$. Here $\pi_j = P_t$ for $j \leq m$ and $1 - P_t$ for $j > m$, with $P_t = \Pr[\mathrm{Bin}(2t, \frac{1}{2}) \geq t] \to \frac{1}{2}$. The coefficients $c_\pi(S)$ are continuous in $\pi \in (0,1)^k$ and Φ is continuous, so both quantities converge to their values at $\pi = \frac{1}{2}$, which are $E_x(\tilde{f}_k)$ and $\mathrm{Inf}_x(\tilde{f}_k) > 0$. The last statement follows from Proposition 11. □

Already $t = 1$, where Th_1 and Th_1' are the OR (disjunction) and the AND of two variables, gives a growing ratio at the forced coordinate, on $2k+1$ variables. By the proof above it is the ratio of $\tilde{f}_k$ under the product measure with bias $\frac{3}{4}$ on $y_1, \dots, y_m$ and $\frac{1}{4}$ on $y_{m+1}, \dots, y_k$. There D and the average over x have coefficients that depend only on $|S \cap \{y_1, \dots, y_m\}|$ and $|S \cap \{y_{m+2}, \dots, y_k\}|$, so the ratio is a double sum over these two numbers and can be computed exactly. It is 1.05, 1.88, 2.57, 3.12, 3.59, 4.12, 4.70 and 5.30 for $k = 3, 5, \dots, 17$, and 13.83, 29.31, 45.28 and 53.32 at $k = 41, 81, 121, 141$, an increase of about 0.39 per unit of k; for $\tilde{f}_k$ the values at these four k are 18.62, 38.11, 57.81 and 67.70. Under a bias the coefficients of the majority near its threshold are Krawtchouk-type sums that change sign, and the proof of Proposition 11 does not carry over. For $t = 1$ the ratio is $(1 + o(1))(k-1)\log_2 \frac{4}{3}$, which is $(1 + o(1))\log_2(1/\mathrm{Inf}_x)$ and so asymptotic to its largest possible value [25]; the proof writes these sums as integrals over the unit circle. The same holds for every fixed $t \geq 1$: the ratio is $(1 + o(1))(k-1)\log_2(1/P_t)$ [26]. At $k = 7$, $t = 1$ the smallest ratio over all coordinates is 2.15, and at $k = 17$, $t = 20$ the forced ratio is 9.59. On monotone functions of at most six variables the rule does no worse than $G_{\mathrm{mon}}(n)$ (Section 7).

n	monotone functions	$G_{\mathrm{mon}}(n)$	$\max H/I$	$\max \sum_i E_i/I$
2	6	2	2	2
3	20	2.672488983973	2.955889582252	2.672488983973
4	168	2.672488983973	3.402475551199	2.704153205629
5	7,581	2.695887520153	3.623740399702	2.883566415773
6	7,828,354	2.944495303872	3.739764017898	3.030964070213
7	2,414,682,040,998	2.954683814183	3.804357069772	3.164564560463

TABLE 1. Maxima over all monotone functions on n variables.

6. The singleton identities do not suffice

For monotone f, $\mu(\{i\}) = \mathrm{Inf}_i(f)^2 = \Pr[i \in U]^2$ for every i, and hence $\sum_i \Pr[i \in U]^2 \le 1$. The following law satisfies both identities, is invariant under permutations, and has unbounded erasure ratio. It is not the spectral distribution of a Boolean function, so it shows only that any proof must use more of the structure.

Proposition 13. *For $n \ge 100$ let ρ be the law of a random subset of $[n]$ containing each element independently with probability $1/n$, conditioned on having at least two elements, let $d = \rho[i \in U]$, and $\lambda = (1 - \sqrt{1-d})/2$. The law μ that gives mass $\rho(S)/4$ to each S with $|S| \ge 2$, mass λ^2 to each singleton and the rest to $\emptyset$ is a probability law with $\mu(\{i\}) = \Pr_\mu[i \in U]^2$, and*

$$\frac{H(\mathbf{1}[i \in U] \mid U \setminus \{i\})}{\Pr_\mu[i \in U]} \;\ge\; \frac{\log n}{288}.$$

Proof. Let $q = \Pr[\mathrm{Bin}(n,1/n) \ge 2] \ge \Pr[\mathrm{Bin}(n,1/n) = 2] = \frac{n-1}{2n}(1-\frac{1}{n})^{n-2} \ge \frac{1}{12}$, using $(1-1/n)^{n-2} \ge e^{-1} > 1/3$. Then $d \le 1/(nq) \le 12/n$, $d/4 \le 3/n < 1/4$, so λ is real, $\lambda \le d/2 \le 6/n$ and $n\lambda^2 \le 36/n \le 0.36$; the mass of $\emptyset$ is at least $1 - \frac{1}{4} - 0.36 > 0$. By the choice of λ, $\Pr_\mu[i \in U] = \lambda^2 + d/4 = \lambda$. In (1) keep only the $T \not\ni i$ with $|T| \ge 2$: both T and $T \cup \{i\}$ lie in the product part, with masses in ratio $(n-1):1$, so each pair contributes $(\mu(T)+\mu(T\cup\{i\}))h_2(1/n)$, and the pairs have total mass $(1/(4q))\Pr[\mathrm{Bin}(n-1,1/n) \ge 2] \ge \frac{1}{4}\cdot\frac{1}{12}$, the last factor by the same estimate as for q. With $\lambda \le 6/n$ and $h_2(1/n) \ge (\log n)/n$ this gives $E_i/\lambda \ge n\, h_2(1/n)/288 \ge (\log n)/288$. □

The exact ratio is 4.59, 5.95, 7.34 and 8.73 at $n = 10^2, 10^3, 10^4, 10^5$.

7. Monotone functions on few variables

Let $G_{\mathrm{mon}}(n)$ be the maximum of $\min_{i:\mathrm{Inf}_i>0} E_i/\mathrm{Inf}_i$ over non-constant monotone functions on n variables, and similarly for H/I and $\sum_i E_i/I$. Monotone functions on n variables are the pairs (f_0, f_1) of monotone functions on $n-1$ variables with $f_0 \le f_1$; there are 7,828,354 of them for $n = 6$ and 2,414,682,040,998 for $n = 7$, the Dedekind numbers. For $n \le 6$ the program `census_mono.py` evaluates every labelled function. The program `mono_census.c` lets the first function of each pair range over one representative of each orbit of the symmetric group on the first $n-1$ variables, weighted by the orbit size, and checks that the weighted count is the Dedekind number. At $n = 6$ its three maxima and its worst cases for the rules of largest and smallest influence agree with those of `census_mono.py` to 10^{-12}, although `mono_census.c` breaks ties in favour of a rule and `census_mono.py` against it. It gives the row $n = 7$. Table 1 gives the results.

For $n = 6$ the maximum of $\min_i E_i/\mathrm{Inf}_i$ is attained by MAJ_3 of three AND_2 on disjoint pairs and by its dual, with every influence $3/16$; the maximum of H/I for $n \le 7$ by AND_n and OR_n.

For $n = 7$ the maximiser of $\min_i E_i/\mathrm{Inf}_i$ has influences 0.09375 at three variables and 0.15625 at four, with ratios 3.0316 and 2.9547; its smallest ratio is at a coordinate of largest influence. For $n \geq 8$ we have only lower bounds. The read-once function $(x_1x_2 \vee x_3x_4) \wedge (x_5x_6 \vee x_7x_8)$ has every influence 21/128 and $E_i/\mathrm{Inf}_i = 3.304701322451$ at every variable, so $G_{\mathrm{mon}}(8) \geq 3.3047$, while $G_{\mathrm{mon}}(7) = 2.9547$; the tribes function [2] $\bigwedge_{j=1}^{5}(x_{2j-1} \vee x_{2j})$, with every influence 81/512, has ratio 3.389192093442 at every variable. The read-once function

$$(x_1 \vee x_2)(x_3 \vee x_4) \vee (x_5 \vee x_6)(x_7 \vee x_8) \vee x_9x_{10}$$

does better: its influences are 49/512 at $x_1, \ldots, x_8$ and 63/512 at x_9, x_{10}, with ratios 3.401069972510 and 3.589953807822, so $G_{\mathrm{mon}}(10) \geq 3.4010$. Both functions are read-once, a class for which the conjecture holds [16, 3]. Up to a permutation of the variables and duality, which leaves every ratio unchanged, a read-once monotone function is a series-reduced planted tree with an OR gate at the root, and there are 261, 2,312 and 68,954 such trees with 8, 10 and 13 leaves [14]. Evaluating all of them for $n \leq 13$ shows that the two functions above are the best read-once functions for $n = 8$ and $n = 10$. For $n = 12$ the best is $x_1x_2 \vee x_3x_4 \vee \cdots \vee x_{11}x_{12}$, the dual of the tribes function with six blocks, with every influence 243/2048 and ratio 3.541677755510 at every variable. Since a function of n variables is also a function of $n + 1$ variables, G_{mon} is nondecreasing, and $G_{\mathrm{mon}}(n) \geq 3.5416$ for $n \geq 12$; for $n = 9$, 11 and 13 the best read-once values, 3.1282, 3.3618 and 3.4242, lie below the bound for $n - 1$. Along the family $\bigvee_{j=1}^{K} x_{2j-1}x_{2j}$, with every influence $\frac{1}{2}(3/4)^{K-1}$, the ratio is 2.3160, 3.0428, 3.3892, 3.5417, 3.5984, 3.6068, 3.5916 and 3.5657 for $K = 3, \ldots, 10$; it is largest at $K = 8$, where it equals 3.606847002096, so $G_{\mathrm{mon}}(n) \geq 3.6068$ for $n \geq 16$. Among ORs of disjoint blocks of the forms x_1x_2, $x_1x_2x_3$, $(x_1 \vee x_2)(x_3 \vee x_4)$ and $(x_1 \vee x_2)(x_3 \vee x_4)(x_5 \vee x_6)$ with at most 20 variables, the best is

$$\bigvee_{b=0}^{2} \bigwedge_{j=1}^{3} (x_{6b+2j-1} \vee x_{6b+2j}),$$

with every influence 12321/131072 and ratio 3.713716018539 at every variable, so $G_{\mathrm{mon}}(n) \geq 3.7137$ for $n \geq 18$. With two blocks of two pairs the same construction is the dual of the eight-variable function above.

Larger ORs of blocks need no truth table. Let $f = \bigvee_b h_b$ with the h_b on disjoint sets of variables, and let ν_b be the indicator that h_b is false. Then $f = -1 + 2\prod_b \nu_b$, so $\hat{f}(T) = 2\prod_b \hat{\nu}_b(T_b)$ for $T \neq \emptyset$, where T_b is the part of T in block b. Since Φ is homogeneous of degree one and $\sum_S \hat{\nu}_b(S)^2 = \Pr[\nu_b = 1]$, for a variable i of block 1, with $\omega = \hat{\nu}_1^2$, $\kappa = \prod_{b>1} \Pr[\nu_b = 1]$ and $\kappa_0 = \prod_{b>1} \hat{\nu}_b(\emptyset)^2$,

$$\mathrm{Inf}_i = 4\kappa \sum_S \omega(S \cup \{i\}),$$

$$E_i = 4\kappa \sum_{S \not\ni i} \Phi\big(\omega(S), \omega(S \cup \{i\})\big) - 4\kappa_0\, \Phi\big(\omega(\emptyset), \omega(\{i\})\big) + \Phi\big(\hat{f}(\emptyset)^2, 4\kappa_0\omega(\{i\})\big).$$

For L copies of $\bigwedge_{t=1}^{J}(x_{2t-1} \vee x_{2t})$ every influence is $\frac{1}{2}(3/4)^{J-1}\big(1-(3/4)^J\big)^{L-1}$. The ratio is largest at $L = 4$ for $J = 3$ ($n = 24$, ratio 3.783969373526), at $L = 6$ for $J = 4$ ($n = 48$, 3.924260765374) and at $L = 8$ for $J = 5$ ($n = 80$, 3.949934743097); this last value is the largest for $J \leq 8$ and $L \leq 40$, and no OR of copies of two of these blocks, at most 30 of each, does better. Hence $G_{\mathrm{mon}}(n) \geq 3.7839$ for $n \geq 24$, $G_{\mathrm{mon}}(n) \geq 3.9242$ for $n \geq 48$ and $G_{\mathrm{mon}}(n) \geq 3.9499$ for $n \geq 80$. Iterated compositions of monotone gates reach 4.506 [22], so the values in the table are far from the supremum over all monotone functions.

n	largest Inf_i	smallest Inf_i	largest $W_i^{(2)}/\mathrm{Inf}_i$	smallest $W_i^{(2)}/\mathrm{Inf}_i$
3	2.672488983973	2.672488983973	2.672488983973	2.672488983973
4	2.672488983973	3.523111966872	3.523111966872	3.523111966872
5	2.695887520153	4.335249356685	4.335249356685	4.335249356685
6	2.944495303872	5.019930166880	4.833940887931	5.019930166880

TABLE 2. Worst ratio at the coordinate selected by each rule, over all monotone functions on n variables.

Table 2 gives, for $n \le 6$, the largest E_i/Inf_i at the coordinate chosen by a rule, with ties broken against the rule. The coordinate of largest influence never does worse than $G_{\mathrm{mon}}(n)$ here, and at $n = 7$ it does not either, with ties broken in its favour. The coordinate of smallest positive influence reaches 6.2425 at $n = 7$ even with ties broken in its favour.

The program `rule_minmax.py` computes $A(f)$, $B(f)$ and every ratio for all monotone functions on $n \le 6$ variables. The largest value over f of $\min_{i \in A \cap B} E_i/\mathrm{Inf}_i$ is 2, 2.672488983973, 2.672488983973, 2.695887520153 and 2.944495303872 for $n = 2, \dots, 6$, which is $G_{\mathrm{mon}}(n)$ in each case. The rule does not always find the best coordinate: the minimum over all coordinates is not attained in $A \cap B$ for 8 of the 168 functions with $n = 4$, 190 of the 7,581 with $n = 5$ and 335,144 of the 7,828,354 with $n = 6$, and among functions whose smallest ratio is positive the quotient of the two minima reaches 1.108, 1.458 and 2.363. At $n = 6$ there are also 360 functions in which some relevant variable has ratio 0 while every variable of $A \cap B$ has a positive ratio, so that the quotient is infinite; for one of them the six ratios are $0, 0, 0.686, 0.686, 0.451, 0.451$ and $A \cap B$ consists of the last two variables. Taking the worst variable of $A \cap B$ instead gives 3.344977967946 at $n = 5$ and 3.650196658624 at $n = 6$; the worst variable of A, of B or of $A \cup B$ gives 4.950362212005 at $n = 6$.

8. Numerical checks

The script `check_hidden.py` in the code repository computes exact Walsh–Hadamard transforms of f_k for odd $k \le 17$, of six substituted functions with blocks of sizes 1, 3 and 5, and of f'_k for even $k \le 18$, and checks Lemma 4, Proposition 6 against a direct evaluation of the erasure entropy at every variable, Corollary 7, the bounds of Theorem 1, each rule of Theorem 2, and Proposition 9 including monotonicity. It also evaluates the law of Proposition 13 exactly. The census programs are `census_mono.py` ($n \le 6$) and `mono_census.c` ($n = 7$). The script `check_small_trees.py` computes the functions on 8 and 10 variables of Section 7 from their truth tables, and `check_table.py` evaluates a function given as a hexadecimal truth table. The script `read_once_census.py` evaluates every read-once monotone function for $n \le 13$, `or_of_and2.py` the family $\bigvee_j x_{2j-1}x_{2j}$ for $K \le 11$, and `or_of_blocks.py` the ORs of blocks with at most 20 variables. The script `or_of_blocks_formula.py` computes larger ORs of blocks by the formula above, after checking it against truth tables of six functions with up to 18 variables. The script `check_xand.py` compares the formula of Proposition 10 with exact transforms of $x \wedge h$ for 300 random monotone h on at most nine variables (largest difference $1.3 \cdot 10^{-15}$) and evaluates it and the bound on a grid of p; `check_gk.py` computes A, B and every ratio of f_k and $\tilde{f}_k$ for odd $k \le 13$ and checks monotonicity. The script `forced_rule.py` builds $F_{k,t}$ for $(k,t) = (3,1), (5,1), (7,1), (9,1), (3,2), (5,2), (3,3)$, checks that $A \cap B = \{x\}$ and that the ratio at x equals the one computed from the biased expansion of $\tilde{f}_k$ (to 10^{-9}), and tabulates the latter for $k \le 17$.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, a project on the use of AI systems in research mathematics [32, 30]. Codex agents running OpenAI's GPT-6.1 found the constructions of Sections 3–6 other than Propositions 10, 11 and 12, with their proofs, and computed the census for $n \leq 6$. Claude agents (Anthropic: Fable and Opus 5.5) computed the census for $n = 7$, found the eight- and ten-variable functions of Section 7, Propositions 10, 11 and 12 and the census of Section 5, checked the proofs and computations, and prepared the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Code and data availability

Code and outputs: `https://github.com/yspennstate/fei-hidden-coordinates`, archived on Zenodo with doi:10.5281/zenodo.24125561.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [30].

Appendix A. Notation

Symbols are listed roughly in the order in which they first appear, and the last column gives that place. The letters i and j index coordinates, s is half the size of an even set in the proof of Proposition 5, b and t index blocks and pairs in the products of Section 7, p and q are also the arguments of Φ, and μ in Proposition 13 is a law on subsets that takes the place of a spectral distribution. The hidden variable x and the variables $x_1, x_2, \ldots$ of Section 7 are different objects. A hat always denotes a Fourier coefficient: $\hat{g}$, $\hat{D}$, $\hat{N}$, $\hat{h}$, $\hat{\varphi}_j$ and $\hat{\nu}_b$ are those of g, D, N, h, φ_j and ν_b.

Symbol	Meaning	First appears
f, n	a Boolean function $\{0,1\}^n \to \{-1,1\}$, and its number of variables	Section 1

Symbol	Meaning	First appears
S, T, $\hat{f}(S)$, χ_S	sets of coordinates, the Fourier coefficient of f at S, and the character $(-1)^{\sum_{j\in S} x_j}$	Section 1
U, μ	the spectral sample, and its law $\mu(S) = \Pr[U = S] = \hat{f}(S)^2$, the spectral distribution	Section 1
$H(\cdot)$, $\mathbb{E}$, Pr	Shannon entropy in bits, with $H(f) = H(U)$ the Fourier entropy; expectation and probability	Section 1
$\mathrm{Inf}_i(f)$, $I(f)$	the influence $\Pr[i \in U]$ of the coordinate i, and the total influence $\mathbb{E}\lvert U\rvert$	Section 1
C	the constant of the Fourier entropy-influence conjecture $H(f) \le C\, I(f)$	Section 1
$E_i(f)$	the erasure entropy $H(\mathbf{1}[i \in U] \mid U \setminus \{i\})$	Section 1
G	$\sup_f \min_{i:\mathrm{Inf}_i(f)>0} E_i(f)/\mathrm{Inf}_i(f)$, finite exactly when the conjecture holds	Section 1
k, MAJ_k	an odd number of inputs (even in Proposition 9), and the majority of k inputs	Section 1
y, y_j, x	a point of $\{0,1\}^k$, its j-th coordinate, and the hidden variable	Section 1
y^0, $[y \ge y^0]$	the point with $(k-1)/2$ ones ($k/2$ in Proposition 9), and the AND of the coordinates at which y^0 is 1	Section 1
f_k	$\mathrm{MAJ}_k(y) \vee (x \wedge [y \ge y^0])$	Section 1
φ_j, F	a non-constant self-dual monotone function of a block of fresh variables, and f_k with each y_j replaced by φ_j	Theorem 1
h_2	the binary entropy function in bits	Section 1
$W_i^{(2)}(f)$	the level-two weight $\sum_{j\ne i} \hat{f}(\{i,j\})^2$	Section 1
$A(f)$, $B(f)$	the variables lying in some shortest minterm, and in some shortest maxterm, of a monotone f	Section 5
$G_{\mathrm{mon}}(n)$	the largest $\min_{i:\mathrm{Inf}_i>0} E_i/\mathrm{Inf}_i$ over non-constant monotone functions on n variables	Section 1
log, ln, e	the logarithms to base 2 and to base e, and the base of the natural logarithm	Section 2
η	$\log_2 e$	Section 2
$\Phi(p,q)$	$p \log \frac{p+q}{p} + q \log \frac{p+q}{q} = (p+q)\, h_2(\frac{q}{p+q})$	Section 2
$\bar{x}$	the complement of a point x of the cube	Section 2
ε	$\Pr[\,\lvert U\rvert$ even$\,]$	Lemma 3
δ, a	2^{-k}, and $\binom{k-1}{(k-1)/2} 2^{1-k}$, the influence of every variable of MAJ_k	Section 3
$\mathbf{1}[\cdot]$	the indicator of an event	Lemma 4
g	the average over x of f_k, of f_k' or of $\tilde{f}_k$	proof of Proposition 5
r, b_r	the size of a set T, and $\lvert\widehat{\mathrm{MAJ}_k}(T)\rvert$ for $\lvert T\rvert = r$ odd	proof of Proposition 5
m	$(k-1)/2$	proof of Proposition 5

Symbol	Meaning	First appears
v, ξ	a uniform point of $\{0,1\}^{2m}$, and a formal variable	proof of Proposition 5
z, z^j, ℓ	the variables of F, those of block j, and one variable of block j	Section 4
ι_j, τ	$\mathrm{Inf}_{y_j}(f_k)$, and $\mathrm{Inf}_\ell(\varphi_j) = \lvert\hat{\varphi}_j(\{\ell\})\rvert$	Section 4
p_T, q_T	$\mu(T)$ and $\mu(T \cup \{y_j\})$ for $T \not\ni y_j$	Proposition 6
S_j, V	the part of S in block j, and a set not containing ℓ	proof of Proposition 6
τ_0, γ_k	$\sqrt{\delta/\iota_j}$, and the bound $\sqrt{2k2^{-k}}\,(k+\eta)$	proof of Theorem 1
w	the block size parameter in $\varphi_j = \mathrm{MAJ}_{2w+1}$	proof of Theorem 2
N, f'_k	the indicator of $\lvert y\rvert \geq k/2$ for even k, and N with the value x at y^0 and $\bar{y}^0$	Section 5
D	the derivative in x of f'_k or of $\tilde{f}_k$	proof of Proposition 9
δ', a_0	2^{1-k}, and $\binom{k}{k/2}2^{-k}$, the influence of every variable of N	proof of Proposition 9
h, p	a monotone function of the variables other than x, and $\Pr[h \text{ true}]$	Proposition 10
$f^\dagger$	the dual of f, its negation at the complement of its argument	Section 5
y^1, M	y^0 with $y^1_{m+1} = 1$, and MAJ_k with the value false at y^1	Section 5
$\tilde{f}_k$	$M(y) \vee (x \wedge [y \geq y^0])$	Section 5
t, Th_t, Th'_t	a positive integer, and the functions of $2t$ variables true when at least t, respectively $t+1$, are true	Section 5
$F_{k,t}$	$\tilde{f}_k$ with y_j replaced by Th_t of a fresh block for $j \leq m$ and by Th'_t for $j > m$	Section 5
Q_j, α_j, β_j	a minterm of φ_j, and the sizes of a shortest minterm and of a shortest maxterm of φ_j ($\alpha_x = \beta_x = 1$)	proof of Proposition 12
Y_0, Y_1	$\{y_1, \ldots, y_m\}$ and $\{y_{m+2}, \ldots, y_k\}$	proof of Proposition 12
π_j, σ_j	the probability that φ_j is true, and $\sqrt{\pi_j(1-\pi_j)}$	proof of Proposition 12
$c_\pi(S)$, ψ_j, R_j	the coefficients of $\tilde{f}_k$ in the basis of the product measure with $\Pr[y_j = 1] = \pi_j$, $(\varphi_j - \pi_j)/\sigma_j$, and a subset of block j	proof of Proposition 12
$\mathrm{Bin}(n,p)$, P_t	a binomial variable, and $\Pr[\mathrm{Bin}(2t, \frac{1}{2}) \geq t]$	proof of Proposition 12
ρ, d, λ	the law of a $\frac{1}{n}$-random subset of $[n]$ conditioned to have two elements or more, $\rho[i \in U]$, and $(1-\sqrt{1-d})/2$	Proposition 13
q	$\Pr[\mathrm{Bin}(n, \frac{1}{n}) \geq 2]$	proof of Proposition 13

Symbol	Meaning	First appears
f_0, f_1	the restrictions of a monotone function to the two values of its last variable	Section 7
$x_1, x_2, \ldots$	the variables of the explicit functions of Section 7	Section 7
K	the number of terms of $\bigvee_{j=1}^{K} x_{2j-1}x_{2j}$	Section 7
h_b, ν_b, T_b	the blocks of an OR of functions on disjoint variables, the indicator that h_b is false, and the part of T in block b	Section 7
ω, κ, κ_0	$\hat{\nu}_1^2$, $\prod_{b>1}\Pr[\nu_b = 1]$ and $\prod_{b>1}\hat{\nu}_b(\emptyset)^2$	Section 7
L, J	the number of copies, and the number of pairs in each copy, of $\bigwedge_{t=1}^{J}(x_{2t-1} \vee x_{2t})$	Section 7

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